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Probability Problem Solving in Data 140

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Contents

Here are summaries of the core problem-solving methods from Data 140, explained with examples.

1

Modeling (Construction/Deconstruction) of Random Variable Distribution

Modeling is understanding. Modeling of random distribution is to tell what a distribution is in terms of the available understandings. In this regard, modeling may also be called expression, expressing concept A in terms of other concepts that have been understood, or better understood.

One of the often powerful expressions is linear combination.

1.1 Randomness Characterizations

(We can't know everything of a random variable's distribution, then we try to get some partial understanding of random distributions, as much as possible.)

Mean the weighted average of all possible values with the weights being the possibilities of the corresponding values. Showing the "center" of the value.

$$E[X - E[X]] = 0$$

Variance Variance measures the "spread" of a random variable around its mean ("center"). Mathematically, it is the expected squared deviation from the mean: (Why squared? indifferent to sign, want it to be non-negative) (Why not use other operation to get non-negative? e.g. absolute value $|X - E[X]|$? We want the transformation/operation to be continuous, to be possible derivation, for example, for finding the maximum/minimum of the variance) (mathematically interesting: the difference between the mean of the square of X and the square of the mean of X)

$$Var(X) = E[(X - E[X])^2]$$

$$Var(X) = E(X^2) - (E[X])^2$$

Simplification

Standard deviation the square root of the variance (Why perform square root operation on the variance? to be more comparable to the random variable's values, and its mean.)

$$SD = \sqrt{Var(X)}$$

Co-Variance

MGF

1.2 Building blocks

1.2.1 Bernoulli distribution (p probability of success)

$P(X = k), k \in \{1, 0\}, P(X = 1) = p$, then $P(X = 0) = 1 - p$, called q

- The simplest random distribution:
 - Only two outcomes (as one outcome wouldn't be random!)
- Express, relate between the outcome of 1 and 0, p to q .
- Mean: $E(X)$: the average of all possible values. Only two possibilities, 1, 0, then the average of them: $1 \times p + 0 \times (1 - p)$ this evaluates to p

Bernoulli $E[X] = p$

- Variance: $Var(X) = E(X^2) - (E[X])^2 = E(X^2) - p^2$ by simplified calculation

Need to figure out $E(X^2)$ (we know $E(X)$ does not automatically implies that we know $E(X^2)$, it definitely $E(X^2) \neq E(X)^2$)

When there is no better means. Try the brutal force of calculating from the definition:

$E(X^2) = \sum_{\text{all possible values of } X^2} p_{X^2} X^2$ where p_{X^2} is the probability of a new random variable $Y, P(Y = X^2)$ as X^2 is a (1-to-1) function, thus $p_{X^2} = P(Y = k^2) = P(X = k) = p$ therefore, $E(X^2) = p1^2 + (1 - p)0^2 = p$ therefore, $Var(X) = E(X^2) - p^2 = p - p^2 = p(1 - p) = pq$

Bernoulli : $E(X^2) = p$

Bernoulli: $Var(X) = pq$

1.3 Using better known distribution as building blocks to study other distributions

The most classic example in Data 140 is the Binomial distribution, which is constructed entirely from Bernoulli trials.

1.3.1 Building the Binomial Distribution

A Binomial (n, p) , n Bernoulli (p) trials. (Repeated Bernoulli experiments)

The Component: Each trial X_i is a Bernoulli random variable where $X_i = 1$ (success) or $X_i = 0$ (failure). The Construction: By adding them together

you create a new distribution that counts the total number of successes in those n trials. Why this works: This “building block” approach allows you to derive the properties of the Binomial distribution directly from the simpler Bernoulli trial. For instance, because expectation is linear, the mean of the Binomial is simply $n \times E[X] = np$.

Binomial: $S_n = \sum_{i=1}^n X_i$
 X_i Bernoulli i.i.d.

Binomial $E[X] = np$

The Probability Mass (Density) Function (PMF) of binomial: – To derive the **Probability Mass Function (PMF)** for the Binomial(n, p) distribution, we view it as a sequence of n independent Bernoulli trials.

1. **1. Probability of a Single Specific Sequence** Imagine a specific sequence of n trials that results in exactly k successes and $n - k$ failures (e.g., $S, S, F, \dots$). Because the trials are independent, we multiply their individual probabilities: The k successes each have probability p , resulting in p^k . The $n - k$ failures each have probability $q = (1 - p)$, resulting in $(1 - p)^{n-k}$. The probability of that one specific sequence is $p^k(1 - p)^{n-k}$.
2. **2. Counting All Possible Sequences** There are many different sequences that can result in k successes. The number of ways to choose exactly k positions for successes out of n available trials is given by the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

3. **3. The Final Density (PMF)** Since each of these $\binom{n}{k}$ sequences is mutually exclusive and has the same probability, we sum them to get the final formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \text{ for } k = 0, 1, \dots, n$$

1.3.2 Building Geometric Distribution

You can also use Bernoulli trials to define “waiting time” distributions: The number of Bernoulli trials needed to get the first success. $P(X = k) = p(1 - p)^{(k-1)}$ where p is the probability of success.

1.3.3 Building Negative Binomial Distribution

The number of Bernoulli trials needed to reach a fixed number of successes.

To derive the Probability Mass Function (PMF) for the negative binomial distribution, we define the random variable X as the total number of trials needed to achieve a fixed number of successes, r .

1. The Logical Requirement For the r -th success to occur exactly on total trial k , two conditions must be met:
 - (a) In the first $k - 1$ trials, there must be exactly $r - 1$ successes.
 - (b) The k -th trial must be a success.

$$\begin{aligned} \text{Binomial: } P(X = k) &= \binom{n}{k} p^k (1 - p)^{n-k} \\ \text{for } k &= 0, 1, \dots, n \\ \binom{n}{k} &= \frac{n!}{k!(n-k)!} \end{aligned}$$

$$\begin{aligned} \text{Geometric distribution (p): } X \\ P(X = k) &= p(1 - p)^{(k-1)} \end{aligned}$$

2. 2. Probability of the First $k - 1$ Trials So the first part follows a Binomial distribution. We need to choose $r - 1$ positions for successes out of $k - 1$ available trials: Ways to arrange: $\binom{k-1}{r-1}$ Probability of $r - 1$ successes: p^{r-1} Probability of $(k - 1) - (r - 1) = k - r$ failures: $(1 - p)^{k-r}$.

This gives us: $\binom{k-1}{r-1} p^{r-1} (1 - p)^{k-r}$.

3. 3. The Final Trial The k -th trial must be a success, which has probability p . Since the trials are independent, we multiply this by our previous result:

$$P(X = k) = \left[\binom{k-1}{r-1} p^{r-1} (1 - p)^{k-r} \right] \times p$$

$$P(X = k) = \binom{k-1}{r-1} p^r (1 - p)^{k-r} \text{ for } k = r, r + 1, \dots$$

4. Alternative Parameterization In some contexts, the distribution is defined by the number of failures (k) observed before the r -th success. In that case, the total trials are $r + k$, and the PMF is written as:

$$P(X = k) = \binom{n+k-1}{k} p^r (1 - p)^k$$

5. Why it is Negative Binomial? The Negative Binomial distribution is named that way primarily because it answers the “inverse” or “negative” version of the question posed by the Binomial distribution.

Here are the two main reasons for the name:

- (a) Inverse Random Variables: In a Binomial distribution, you fix the number of trials and count the random number of successes. In a Negative Binomial distribution, you do the opposite: you fix the number of successes you want to achieve and count the random number of trials it takes to get there.

Negative Binomial: X

$P(X = k) =$

$\binom{k-1}{r-1} p^r (1 - p)^{k-r}$

for $k = r, r + 1, \dots$

k : the number of total trials

Two formulations of negative binomial (geometric):

1. (k : total number of trials to achieve fixed number r success)

2. (k : number of failures to achieve fixed number r success)

geometric: is negative binomial with $r = 1$ constant omitted

Distribution	Fixed Quantity	Random Variable	Interpretation	Relationships
Binomial	# of trials (n)	# of successes (k)	Count successes in a fixed # of trials	$n \geq k$
Negative Binomial	# of successes (r)	# of trials (k)	Count trials needed to reach a fixed # of successes	$k \geq r$
Negative Binomial	# of successes (r)	or # of failures (k)	Count failures needed to reach a fixed # of successes	$k + r = \text{total trials}$

Both distributions assume independent Bernoulli trials with constant p

Practical modeling insight

Use Binomial when your experiment budget is fixed. Use Negative Binomial when your stopping condition is success-based (common in reliability testing, queuing, and retry systems).

If you want, I can map this to likelihood functions and show how they differ in inference workflows.

1.3.4 Negative Binomial degenerates (collapses) to Geometric distribution

To show that the Geometric distribution is a special case of the Negative Binomial distribution, we simply set the required number of successes (r) to 1 in the Probability Mass Function (PMF).

1. The Mathematical Substitution

(a) Negative Binomial PMF: This distribution models the number of trials (k) to get r successes:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

(b) Set $r = 1$:

$$P(X = k) = \binom{k-1}{1-1} p^1 (1-p)^{k-1}$$

(c) Simplify:

- $\binom{k-1}{0} = 1$ (There is only one way to have the first success on the k -th trial: $k-1$ failures followed by one success).
- Result: $P(X = k) = p(1-p)^{k-1}$.

This final expression is exactly the PMF for the Geometric distribution, defined as the number of trials until the first success.

2. Conceptual Link

- Negative Binomial: “How many trials until I get r successes?”.
- Geometric: “How many trials until I get the first success?”.

When $r = 1$, the questions are identical.

1.3.5 Law of Small Numbers: Poisson as approximation of the Binomial distribution

“rare events” where the number of trials (n) is large and the probability of success (p) is very small, such that np converges to a constant λ .

Detail development: The **Law of Small Numbers** describes how the Binomial distribution converges to the Poisson distribution when the number of trials n is very large and the probability of success p is very small, such that their product $np = \lambda$ remains constant.

1. **Mathematical Derivation** To derive this, we start with a variable $X_n \sim \text{Binomial}(n, \lambda/n)$ and take the limit as $n \rightarrow \infty$:

(a) Set up the Binomial PMF:

$$P(X_n = k) = \frac{n(n-1) \dots (n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

(b) Rearrange the terms:

$$P(X_n = k) = \frac{\lambda^k}{k!} \left[\frac{n}{n} \cdot \frac{n-1}{n} \dots \frac{n-k+1}{n} \right] \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}$$

(c) Evaluate the limits as $n \rightarrow \infty$:

- The product in the brackets $\left[\frac{n}{n} \cdot \frac{n-1}{n} \dots \frac{n-k+1}{n}\right]$ consists of k terms that each approach 1.
- The term $\left(1 - \frac{\lambda}{n}\right)^n$ approaches $e^{-\lambda}$.
- The term $\left(1 - \frac{\lambda}{n}\right)^{-k}$ approaches 1 because k is a fixed constant while n grows.

(d) Final Result: By combining these limits, we arrive at the Poisson PMF:

$$\lim_{n \rightarrow \infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

This approximation is considered accurate in practice when n is large and the average λ is relatively small.

2. **Poisson Distribution Definition** A random variable X follows a Poisson (λ) distribution if its Probability Mass Function (PMF) is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \text{ for } k = 0, 1, 2, \dots$$

where $\lambda > 0$ is the average rate of occurrence.

The Poisson distribution was developed as a mathematical limit of the **Binomial distribution**, discovered by Siméon Denis Poisson in 1837.

Rare event of Binomial X :
 $\lim_{n \rightarrow \infty} p \rightarrow 0 \quad np = \lambda$
 for Binomial (n, p) $X \quad E[X] = np$
 for Binomial X_n
 $\lim_{n \rightarrow \infty} P(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

Poisson X : $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$
 for $k = 0, 1, 2, \dots$
 $\lambda > 0$ the average rate of occurrence

- (a) **The Core Intuition: “The Law of Small Numbers”** The primary intuition source is the study of **rare events** occurring within a large number of independent trials. **The Limit:** If you have a sequence of trials (n) that goes to infinity while the probability of success (p) goes to zero, but their product ($np = \lambda$) stays constant, the distribution converges to Poisson. **The “Building Block”:** It essentially treats a continuous region of time or space as being divided into many tiny “bins” where a “success” (an event) is extremely unlikely in any single bin.
- (b) **Historical Real-World Sources** The distribution became famous through observations of phenomena that fit this “rare event” model perfectly: **Radioactivity:** Counting the clicks on a Geiger counter over time. **Military Data:** The number of Prussian soldiers killed by horse kicks annually across different army corps. **Wartime Clustering:** The hits from Nazi V1 “Buzz Bombs” in London during WWII, where the city was divided into small sectors to see if the bombs were clustering or falling randomly.
- (c) **Key Logical Assumptions** For this intuition to hold, four things must be true:
- Independence:** One event does not affect the likelihood of another.
 - Constant Rate:** Events occur at a steady average frequency (λ).
 - Proportionality:** The number of events is proportional to the size of the interval.
 - Non-Simultaneity:** Two events cannot happen at the exact same instant or location.

Poisson: rare events of arrival to some “bins” over (long) time or space with average rate of λ

1.3.6 Logarithmic Series distribution (not required in Data 140)

The Logarithmic Series distribution is a discrete probability distribution used to model counts where the value zero is not possible, meaning its support is $k = 1, 2, 3, \dots$

1. Probability Mass Function (PMF) As a member of the **power series family**, its PMF is defined by the parameter θ ($0 < \theta < 1$):

$$P(X = k) = \frac{\theta^k}{-k \ln(1 - \theta)}$$

In this structure, the “coefficient” is $a(k) = 1/k$ and the “series function” is $f(\theta) = -\ln(1 - \theta)$.

2. Meaning and Intuition

- **Species Abundance:** It is most famous for modeling biological diversity, such as the number of individual insects found for each species in a collection.
- **Zero-Truncated Limit:** Intuitively, you can think of it as the limiting form of a Negative Binomial distribution that has been “zero-truncated” (the $k = 0$ case is removed).
- **Long-Tailed Behavior:** It is a “long-tailed” distribution, meaning it can model scenarios where most categories have very few members, but a few rare categories have very high counts.

3. Visual Diagnosis In an Ord Plot, the Logarithmic Series is identified by a positive slope ($b > 0$) and a negative intercept ($a < 0$).

1.3.7 Power series family

The **power series family** is a broad group of discrete probability distributions where the probability of a value k is defined by the formula $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$.

In this structure, θ is a positive parameter, $a(k)$ is a coefficient function of k , and $f(\theta)$ is the “series function” that ensures the probabilities sum to one.

Power series family:

$$P(X = k) = \frac{a(k)\theta^k}{f(\theta)}, \theta > 0$$

1. Log-Ratio of Power Series Distributions Given $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$, then $R(k) = \frac{P(k)}{P(k-1)} = \frac{a(k)}{a(k-1)}\theta$ significantly simplified!

ratio of consecutive probabilities:

$$\frac{P(k)}{P(k-1)} = \frac{a(k)}{a(k-1)}\theta$$

2. Components of Power Series Distributions

Distribution	$P(X = k)$	Coefficient ($a(k)$)	Parameter (θ)	Series Function ($f(\theta)$)
Poisson (k : # of successes out of infinity)	$\frac{e^{-\lambda}\lambda^k}{k!}$	$1/k!$	λ	e^θ
Binomial (k : # of successes of n trials)	$\binom{n}{k}p^k(1-p)^{n-k}$	$\binom{n}{k}$	$p/(1-p)$	$(1+\theta)^n$
Negative Binomial (k : total # of trials to achieve fixed # r success)	$\binom{k-1}{r-1}p^r(1-p)^{k-r}$	$\binom{k-1}{r-1}$	$1-p$	$\left(\frac{\theta}{(1-\theta)}\right)^r$
Negative Binomial (k : # of failures to achieve fixed # r success)	$\binom{r+k-1}{r-1}p^r(1-p)^k$	$\binom{r+k-1}{r-1}$	$1-p$	$(1-\theta)^{-r}$
Geometric (k : # of total trials till success)	$p(1-p)^{(k-1)}$	1	$1-p$	$\frac{\theta}{(1-\theta)}$
Geometric (k : # of failures till success)	$p(1-p)^{(k)}$	1	$1-p$	$(1-\theta)^{-1}$
Logarithmic Series	$\frac{\theta^k}{-k \ln(1-\theta)}$	$1/k$	θ	$-\ln(1-\theta)$

More compontes of the power series distributions:

Distribution

Poisson (k : # of successes out of infinity)Binomial (k : # of successes of n trials)Negative Binomial (k : total # of trials to achieve fixed # r success)Negative Binomial (k : # of failures to achieve fixed # r success)Geometric (k : # of total trials till success)Geometric (k : # of failures till success)

Logarithmic Series

$$R(k) = \frac{P(k)}{P(k-1)} = \frac{a(k)}{a(k-1)}\theta$$

$$\frac{\theta}{k} = \frac{\lambda}{k}$$

$$R(k) = \frac{n-k+1}{k} \cdot \frac{p}{1-p}$$

$$\frac{k-1}{k-r}(1-p)$$

$$\text{or let } k' = k - r, R(k') = \frac{k'+r-1}{k'}(1-p)$$

In fact, k' is just the # of failures!

$$R(k) = \frac{k+r-1}{k} \cdot (1-p)$$

$$\theta = 1 - p$$

$$\theta = 1 - p$$

$$\frac{k-1}{k}\theta$$

$$\text{Linear Recurrence } kR(k) = ak + b$$

$$\lambda = 0k + \lambda$$

$$\left(\frac{-p}{1-p}\right)k + \frac{p(n+1)}{1-p}$$

(hard to convert to $ak + b$ form,use the equivalent with k as # of failures)

$$(1-p)k' + (r-1)(1-p)$$

$$(1-p)k + (r-1)(1-p)$$

$$(1-p)k$$

$$(1-p)k$$

$$\theta k - \theta$$

3. identifying the components of a power series distribution $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$:

(a) Overall receipt:

- Identify $a(k)$: Isolate the coefficient function that depends on k but is not part of the term with k as an exponent.
- Identify θ^k : Group all terms where k is the power; the base of this expression is your parameter θ .
- Determine $f(\theta)$: The remaining terms (which do not depend on k) represent $\frac{1}{f(\theta)}$. Take their reciprocal to find the series function $f(\theta)$.

(b) Poisson In $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$:

- $a(k) = 1/k!$ leaving $e^{-\lambda}\lambda^k$
- $\theta = \lambda$ leaving $e^{-\lambda}$
- $1/f(\theta) = e^{-\lambda} \Rightarrow f(\theta) = e^\theta$.

(c) Binomial

$$\text{In } \binom{n}{k}p^k(1-p)^{n-k}$$

- $a(k) = \binom{n}{k}$ leaving $p^k(1-p)^{n-k}$ rearranging $p^k(1-p)^{n-k} = \left[\frac{p}{1-p}\right]^k (1-p)^n$
- $\theta = \frac{p}{1-p}$ leaving $(1-p)^n$
- $1/f(\theta) = (1-p)^n$ needing to express $1-p$ in terms of $\theta = \frac{p}{1-p}$

Trying to construct relationship between $1-p$ and θ .

We have $p + (1-p) = 1$ creating the appearance of θ in the equation, by deviding $1-p$ at the both side of the above equation, resulting another equation:

$$\frac{p}{1-p} + 1 = \frac{1}{1-p}$$

$$\begin{aligned} \text{tricks: with } \theta &= \frac{p}{1-p} \\ \text{use } p + (1-p) &= 1 \\ \text{to get } \frac{p}{1-p} + 1 &= \frac{1}{1-p} \\ \theta + 1 &= \frac{1}{1-p} \end{aligned}$$

thus we have $\theta + 1 = \frac{1}{1-p}$ therefore, we have $1 - p = \frac{1}{\theta+1}$

thus

$$1/f(\theta) = (1-p)^n = \frac{1}{(\theta+1)^n}$$

Conclusion: $f(\theta) = (\theta+1)^n$

(d) Negative binomial

In $\binom{k-1}{r-1} p^r (1-p)^{k-r}$ (k : total # of trials to achieve fixed # r success)

i. $a(k) = \binom{k-1}{r-1}$ leaving $p^r (1-p)^{k-r}$ rearranging $p^r (1-p)^{k-r} = (1-p)^k p^r (1-p)^{-r} = (1-p)^k \left(\frac{p}{1-p}\right)^r$

ii. $\theta = 1-p$ leaving $\left(\frac{p}{1-p}\right)^r = \left(\frac{1-\theta}{\theta}\right)^r$

iii. $1/f(\theta) = \left(\frac{1-\theta}{\theta}\right)^r$ therefore $f(\theta) = \left(\frac{\theta}{1-\theta}\right)^r$

In $\binom{r+k-1}{r-1} p^r (1-p)^k$ (k : # of failures to achieve fixed # r success)

i. $a(k) = \binom{r+k-1}{r-1}$ leaving $p^r (1-p)^k$

ii. $\theta = 1-p$ leaving p^r as $p = 1-\theta$

iii. $1/f(\theta) = (1-\theta)^r$ therefore $f(\theta) = (1-\theta)^{-r}$

(e) Geometric

For k to be the number of trials to get a success $k = 1, 2, \dots$ In $p(1-p)^{(k-1)}$

i. $a(k) = 1$ remaining $p(1-p)^{(k-1)}$

ii. $\theta = 1-p$ remaining $p(1-p)^{-1}$

iii. $\frac{1}{f(\theta)} = p(1-p)^{-1}$ expressing $p = 1-\theta$, then $p(1-p)^{-1} = \frac{(1-\theta)}{\theta}$ therefore, $f(\theta) = \frac{\theta}{(1-\theta)}$

For k to be the number of failurs before a success $k = 0, 1, 2, \dots$ In $p(1-p)^{(k)}$

i. $a(k) = 1$ remaining $p(1-p)^{(k)}$

ii. $\theta = 1-p$ remaining p

iii. $\frac{1}{f(\theta)} = p$ expressing $p = 1-\theta$, then therefore, $f(\theta) = (1-\theta)^{-1}$

(f) Logarithmic Series

In $\frac{\theta^k}{-k \ln(1-\theta)}$

i. $a(k) = \frac{1}{k}$ leaving $\frac{\theta^k}{-\ln(1-\theta)}$

ii. $\theta = \theta$ leaving $\frac{1}{-\ln(1-\theta)}$

iii. $\frac{1}{f(\theta)} = \frac{1}{-\ln(1-\theta)}$ therefore, $f(\theta) = -\ln(1-\theta)$

4. Application of Power Series Distributions The power series relationships are particularly useful because they allow us to linearize the probability mass function for graphical diagnosis.

A fundamental property of this family is that the log-ratios of consecutive probabilities follow a linear relationship when plotted against k . This makes the family particularly easy to identify using diagnostic tools like Ord plots or Poissonness plots, where the slope and intercept determine which specific distribution fits the data.

The idea of linearizing the probability mass function (PMF) through log-ratios is a powerful diagnostic tool because it transforms complex, non-linear relationships (involving powers and factorials) into a simple straight line for visual inspection.

- (a) 1. Why use Log-Ratios? For members of the power series family, the PMF is defined as $P(X = k) = \frac{a(k)\theta^k}{f(\theta)}$. When you take the ratio of consecutive probabilities, $\frac{P(k)}{P(k-1)}$, the common “series function” $f(\theta)$ cancels out, simplifying the expression significantly. Taking the logarithm then linearizes the parameters, often isolating the parameter θ as a linear component.

- (b) 2. Diagnostic Tools Ord Plot: This tool plots the relationship $k \frac{P(k)}{P(k-1)} = ak + b$. If the resulting plot is linear, the data belong to the Katz family. The specific distribution is identified by the signs of the slope (a) and intercept (b):

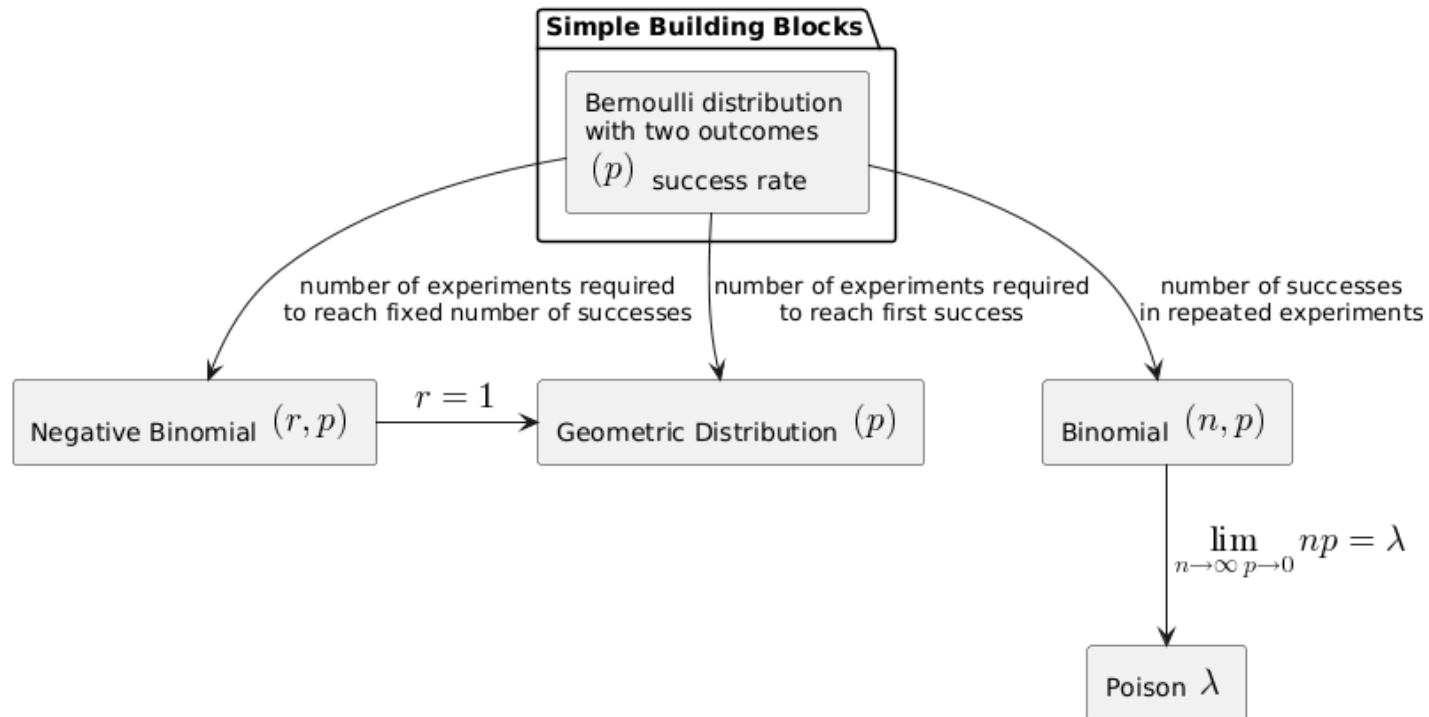
- Poisson: Slope is zero.
- Binomial: Slope is negative.
- Negative Binomial: Slope is positive and intercept is positive.
- Logarithmic Series: Slope is positive and intercept is negative.

Poissonness Plot: This more robust method uses a “count metameter”—a log-transformation of observed frequencies, $k!$, and sample size—designed to produce a straight line specifically if the data follow a Poisson distribution.

- (c) 3. Mathematical Intuition Linearization allows you to move from histograms, which can be hard to compare visually against curvilinear models, to a horizontal or linear reference. In a leveled plot, for instance, you can judge deviations from a horizontal line at zero, making it much easier to spot outliers or lack of fit.

NEXT:

1.3.8 “Family Tree” of Distributions



2

Foundational Probability Calculations

2.1 Start from definitions

Perform reasoning and calculation definitions. This is often is the more straightforward approach that we don't use enough.

For example, the calculation of $E(X^2)$ where X is Bernoulli

$$E(X^2) = \sum_{\text{all possible values of } X^2} p_{X^2} X^2$$

2.2 Function application to random variable

$P(X = k) = P(f(X) = f(k))$ When you apply a 1-to-1 (injective) function f to a discrete random variable X , the probability of the new variable $Y = f(X)$ taking the value $f(k)$ is identical to the probability of X taking the value k .

2.2.1 Why this works

Because the mapping is 1-to-1, the event $\{X = k\}$ is logically equivalent to the event $\{f(X) = f(k)\}$. Since they represent the same set of outcomes in the sample space, they must have the same probability:

$$P(X = k) = P(f(X) = f(k))$$

2.2.2 Examples from our discussion

1. Squaring a Bernoulli: We previously saw that for a Bernoulli variable $X \in \{0, 1\}$, applying $f(X) = X^2$ is a 1-to-1 mapping on its domain. Therefore, $P(X^2 = 1) = P(X = 1) = p$ and $P(X^2 = 0) = P(X = 0) = 1 - p$

2. Monotone Transformations: In Data 140, we often use monotone functions (which are always 1-to-1) to transform variables. a function is not 1-to-1 ($f(X) = X^2$ for a variable that can be both positive and negative) and how that changes the probability? For example, if $X \in \{-1, 0, 1\}$, then $f(X) \in \{0, 1\}$ (3 elements in input domain to 2 elements in the outputs, both 1 and -1 mapped to 1, that is, the probability of $P(f(x) = 1) = P(X = -1) + P(X = 1)$)

2.3 Linearity of Expectations:

The expected value of a sum is the sum of the individual expectations, regardless of whether the variables are independent.

$$E \left[\sum_{i=1}^n \lambda_i X_i \right] = \sum_{i=1}^n \lambda_i E[X_i]$$

2.3.1 Applicaiton Example: Why is $Var(X) = E(X^2) - (E[X])^2$?

By definition: $Var(X) = E[(X - E[X])^2]$

By algebra expression (to figure out how to compute $(X - E[X])^2$ first, by $(a + b)^2 = (a - b)(a - b) = a^2 - 2ab - b^2$) $(X - E[X])^2 = X^2 - 2XE[X] + E[X]^2$

By linearity of expectation: $E[X^2 - 2XE[X] + E[X]^2] = E[X^2] - E[2XE[X]] + E[E[X]^2]$ For mean with respect to X , both $E[X]$ are constants, and $E[constant] = constant$ therefore $E[E[X]^2] = E[X]^2$ and $E[2XE[X]] = (2E[X])E[X] = 2E[X]^2$

By algeric substitution (replacing the equal value terms with simpler forms): $E[X^2 - 2XE[X] + E[X]^2] = E[X^2] - 2E[X]^2 + 2E[X]^2 = E[X^2] - E[X]^2$

2.3.2 Applicaiton Example: The mean of a Binomial(n, p) distribution is np because it is the sum of n Bernoulli(p) trials, each with mean p .

2.3.3 Variance and Covariance Rules: The variance of a sum includes the sum of individual variances plus twice the covariances of all pairs.

1. 1. Proof: Variance of a Sum The variance of a sum of random variables is derived from the definition of variance: $Var(X) = E[(X - E[X])^2]$

For a sum $S = \sum_{i=1}^n X_i$:

(a) Linearity of Expectation: $E[S] = \sum E[X_i]$.

(b) Substitution: $Var(S) = E \left[\left(\sum (X_i - E[X_i]) \right)^2 \right]$ and let $(X_i - E[X_i]) = a_i$, then $Var(S) = E \left[\left(\sum a_i \right)^2 \right]$.

(c) Expansion: Expanding the square of a sum $(\sum a_i)^2$ gives $\sum a_i^2 + 2 \sum_{i < j} a_i a_j$ (very elegant and convenient expression!), thus
 $Var(S) = E \left[\sum a_i^2 + 2 \sum_{i < j} a_i a_j \right]$.

(d) Expectation of Terms:

- The terms $E[(X_i - E[X_i])^2]$ are the individual variances, $\sum Var(X_i)$.
- The cross-product terms $E[(X_i - E[X_i])(X_j - E[X_j])]$ are the covariances, $Cov(X_i, X_j)$.

Result: $Var(\sum X_i) = \sum Var(X_i) + 2 \sum_{i < j} Cov(X_i, X_j)$.

If the variables are independent, all covariance terms are zero, simplifying the formula to $Var(\sum X_i) = \sum Var(X_i)$.

2.4 MGF for Distribution Identification: The Moment Generating Function (MGF) of the sum of independent variables is the product of their individual MGFs.

Example: Multiplying the MGFs of two independent Poisson variables (μ and λ) results in the MGF for a Poisson($\mu + \lambda$).

2.5 Taylor Series Transformations: Using the expansion $e^z = \sum z^k / k!$ to simplify complex expectations into recognizable functional forms.

Example: This was used to simplify the expectation $E[e^{tX}]$ to derive the Poisson MGF $e^{\lambda(e^t - 1)}$.

3

Variance and Co-Variance

4

Optimization and Scaling

1. **Log-Transformations:** Converting products into sums to simplify differentiation, which is essential for Maximum Likelihood (MLE) and MAP estimates [History].
 - **Example:** Taking the natural log of $x^6(1-x)^4$ to find the mode of a Beta distribution via simpler addition-based derivatives [History].
2. **Derivative-Based Optimization:** Setting the first derivative of a density or log-density to zero to find the parameter value that maximizes the probability [History].
 - **Example:** Differentiating the log-posterior $\frac{6}{x} - \frac{4}{1-x} = 0$ to find the MAP estimate of 0.6 [History].
3. **Monotonic Function Application:** Since $\ln(x)$ is a monotonic increasing function, the value of x that maximizes $\ln(f(x))$ is guaranteed to be the same value that maximizes $f(x)$ [History].
 - **Example:** This allows us to ignore complex constants in the Beta density when finding the MAP [History].

5

Discrete and Bayesian Inference

1. **Recursive Odds Ratios:** Calculating $P(k)$ using the ratio $R(k) = \frac{P(k)}{P(k-1)}$, which avoids computationally heavy factorials for large n .
 - **Example:** To find $P(36)$ in a Binomial(60, 2/3), multiply $P(35)$ by the simple ratio $\frac{61-36}{36} \cdot 2$.
2. **Discrete Mode Identification:** Finding the peak of a histogram by identifying where the odds ratio $R(k)$ drops below 1.
 - **Example:** The mode of a Binomial(n, p) is the integer part of $(n + 1)p$.
3. **Conjugate Prior Updating:** Using the “Beta-Binomial” partnership to update beliefs by simply adding successes and failures to the prior parameters [22, History].
 - **Example:** A Beta(2, 3) prior combined with 2 heads in 7 tosses instantly becomes a Beta(4, 8) posterior [History].
4. **Order Statistics Mapping:** (Poisson Approximation)** **Method:** When n is very large and p is very small, you can approximate Binomial probabilities using the **Poisson** (λ) distribution, where $\lambda = np$. **Example:** Calculating the probability of a rare disease affecting exactly 5 people in a city of 100,000. Instead of a massive Binomial calculation, you use the Poisson PMF with $\lambda = 100,000 \cdot p$.

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15. Poissonization

Method: A technique where you treat the total number of trials n as a random variable following a Poisson distribution. **Why it works:** This often “decouples” variables that are normally dependent. For example, if the number of people entering a store is Poisson, the number of men and women entering become **independent** Poisson variables.

16. Markov Chain Steady-State Analysis

Method: Modeling systems that transition between states where the next step depends only on the current state. You find the long-run “steady state” by solving the matrix equation $\pi P = \pi$. **Example:** Predicting the long-term market share of two competing brands based on monthly customer switching probabilities.

17. Central Limit Theorem (CLT) Approximation

Method: For a large sample size n , the distribution of the sample sum (or mean) is approximately **Normal**, regardless of the original distribution's shape. **Example:** Estimating the probability that the average height in a sample of 200 people is within a certain range using the Z -table.

18. Beta-Binomial Probability (The “C-Constant” Formula)

Method: Calculating the probability of k successes in n trials when the success probability p is unknown and follows a Beta(r, s) distribution. **Formula:** $P(S_n = k) = \binom{n}{k} \frac{C(r,s)}{C(r+k, s+n-k)}$ where C is the normalizing constant of the Beta density.

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19. Multiple Linear Regression (Matrix Form)

Method: Finding the best-fit parameters β for a model with multiple predictors using the matrix identity $\hat{\beta} = (X^T X)^{-1} X^T Y$. **Example:** Predicting a house price (Y) based on square footage, age, and location (X_1, X_2, X_3) simultaneously.

11

Other pending topics to discuss

11.1 Covariance: $Cov(X_i, X_j) = E[(X_i - E[X_i])(X_j - E[X_j])]$

11.2 Unbiased Estimator:

When working with sample data, we use the divisor $(n - 1)$ in the formula $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ to ensure the estimate of σ^2 is unbiased.

11.2.1 2. Variance vs. Sample Variance

While the proof above describes the theoretical Variance (σ^2), statisticians often work with Sample Variance (S^2).

Feature	Variance (σ^2)	Sample Variance (S^2)
⋮	⋮	⋮
Nature	A fixed parameter of a population or distribution.	A statistic calculated from a finite sample of data.
Calculation	$E[(X - \mu)^2]$	$\frac{1}{n-1} \sum (X_i - \bar{X})^2$.
Goal	Describes the true underlying spread.	Estimates the true population variance.

1. 3. Why use $(n - 1)$? (Unbiased Estimation) The sample variance S^2 is an unbiased estimator of σ^2 because its expected value equals the true population variance: $E[S^2] = \sigma^2$.

The “Bias” Problem: Using the population mean μ would allow a divisor of n . However, we use the sample mean $\bar{X}$, which is already “centered” in the data. This makes the deviations $(X_i - \bar{X})$ smaller on average than $(X_i - \mu)$. The Correction: Dividing by $n - 1$ (Bessel’s correction) perfectly compensates for this underestimation, ensuring the estimate is accurate on average.

11.3 Independent random variables

Two random variables X and Y are mathematically **independent** if the probability of any joint outcome is simply the product of their individual probabilities.

11.3.1 Mathematical Definitions

Discrete Variables: X and Y are independent if, for all possible values x and y :

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$$

. Continuous Variables: Independence occurs when the joint probability density function (PDF) is the product of the individual marginal densities:

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$$

.

11.3.2 Key Properties in Data 140

1. **Expectation:** For independent variables, the expectation of their product equals the product of their expectations: $E[XY] = E[X]E[Y]$.
2. **Variance:** The variance of a sum of independent variables is simply the sum of their variances ($Var(X + Y) = Var(X) + Var(Y)$) because their **covariance is zero** [History].
3. **MGFs:** The Moment Generating Function of a sum of independent variables is the product of their individual MGFs: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$ [History].
4. **Conditional Probability:** If X and Y are independent, knowing the value of Y does not change the distribution of X : $P(X | Y) = P(X)$ [History].

In the **Bayesian framework** we discussed, we often move from assuming trials are “independent” to viewing them as **exchangeable**, where future trials depend on the evidence gathered from previous ones [History].

Would you like to see how this definition of independence is used to prove that the sum of independent **Normal** variables is also Normal?

Appendix: Hints of remembering the number of combinations formula

 $\binom{n}{k}$

Use the following equation:

$$n! = \binom{n}{k} k! (n - k)!$$

The total permualtions of n elements can be achieved by all possible k permutations and $(n - k)$ permutations.

$$n! = \binom{n}{k} k! (n - k)!$$

13

Core skills of mathematics required for probability

13.1 Calculus

1. Limit operation
2. Derivation
3. Integral
4. Function expression as series
 - (a) Taylor series
5. e

13.2 Linear algebra

13.3 Combinatory

13.4 Trigonometry

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*Would you like to see how these values are used to construct an **Ord Plot** to identify a distribution from raw data?*